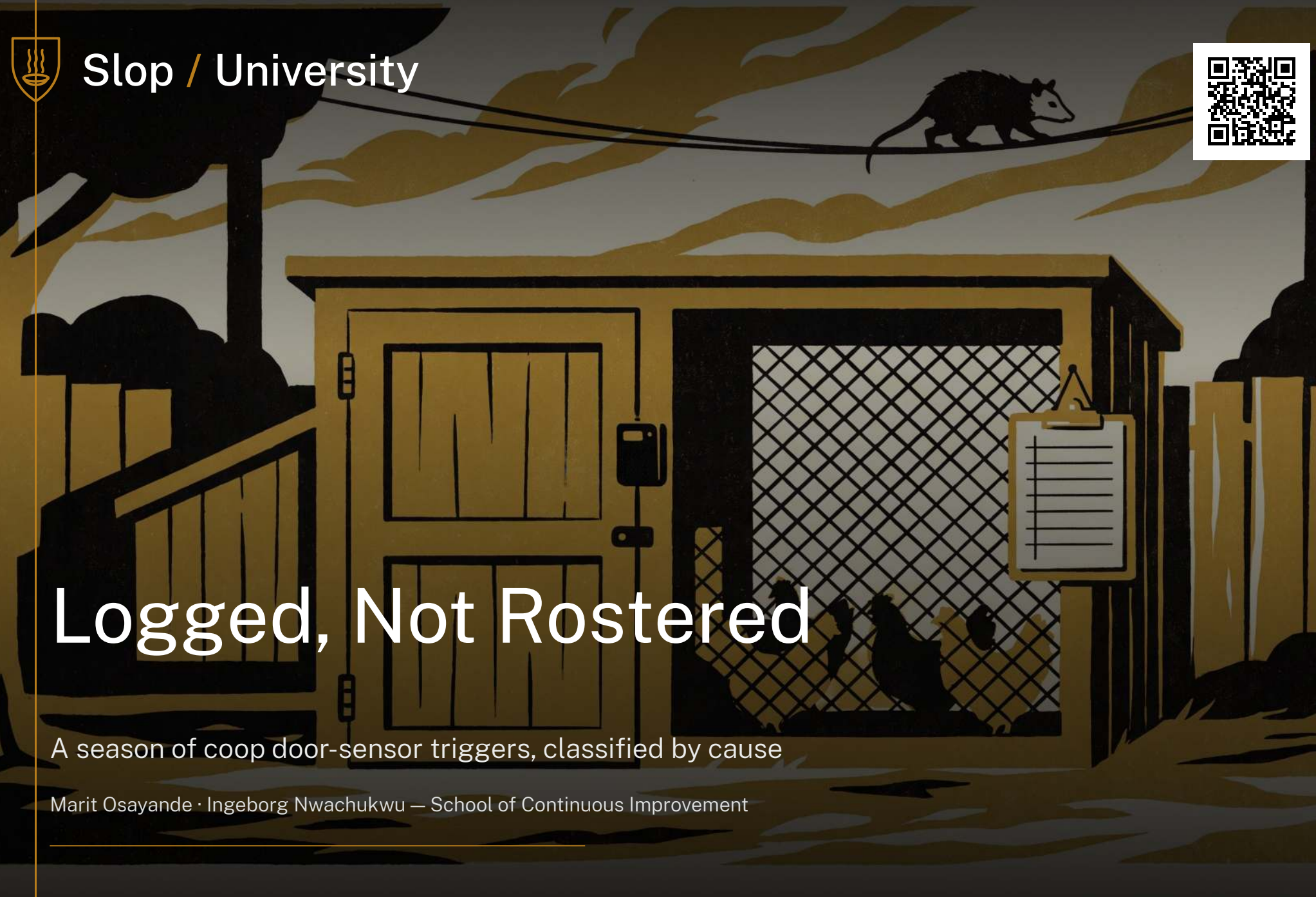




Logged, Not Rostered

A season of coop door-sensor triggers, classified by cause

Marit Osayande · Ingeborg Nwachukwu — School of Continuous Improvement

Background

A street’s shared coop keeps its roster on a clipboard and its trust on the honour system. Once a door sensor started logging every open and close, what did that record actually contain?

Aims

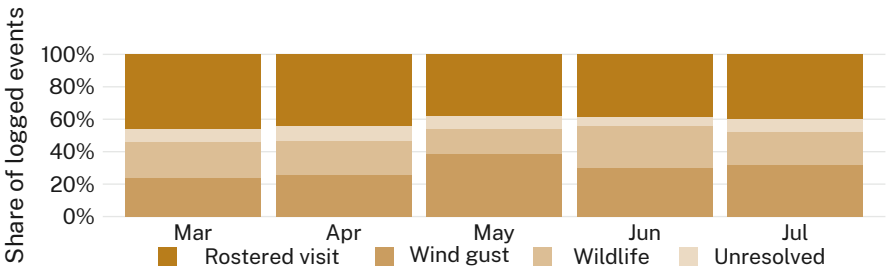
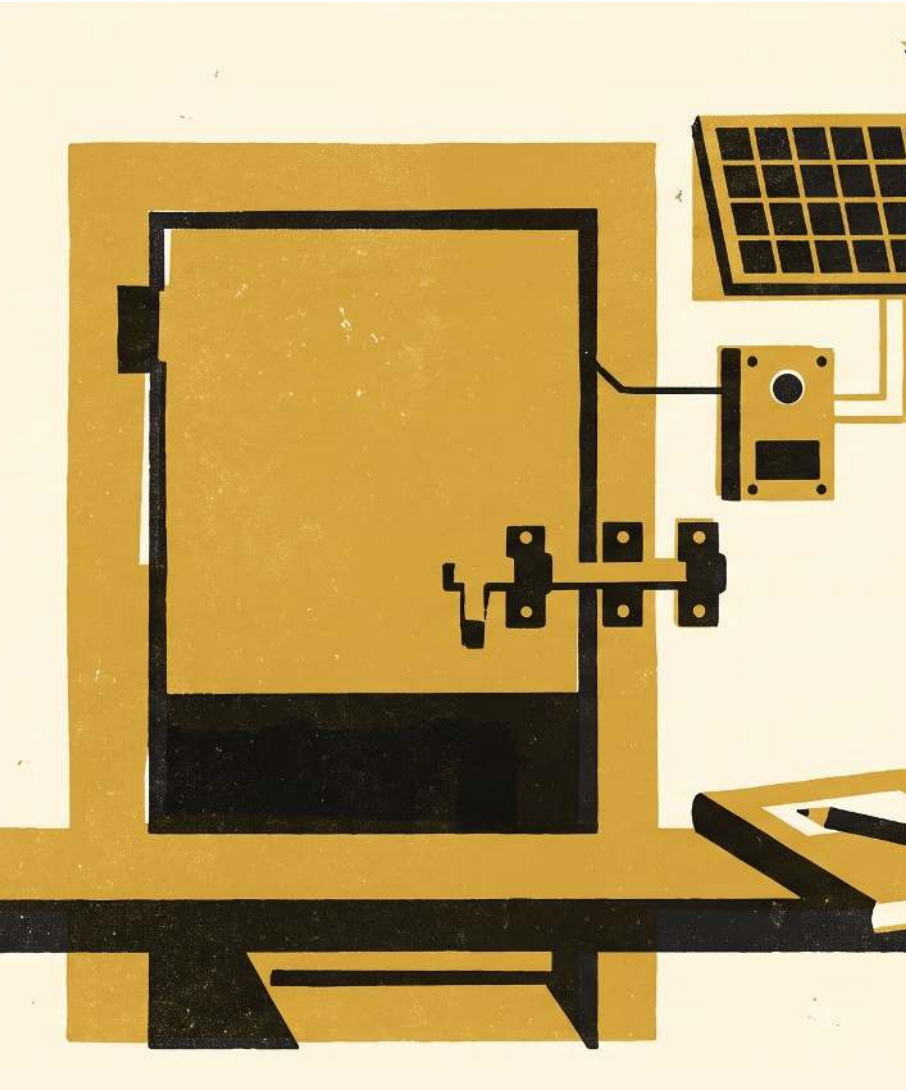
- Characterise a season of raw door-sensor events beyond the duty visits it was installed to confirm
- Classify each event against the published roster, a resident’s possum log, and the local wind-gust record
- Track how the mix of causes shifts across the season

Methods

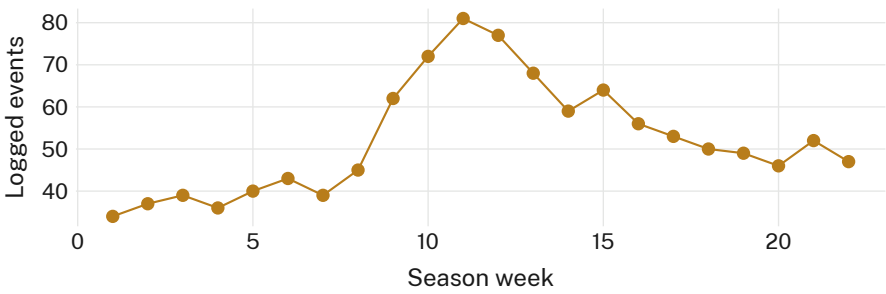
- 1 coop · reed-switch door sensor + PIR motion logger · 6 co-owning households · 22-week season (March-July)
- Each event timestamped, cross-checked against the paper duty roster, one household’s possum-sighting log, and the Bureau’s local wind-gust record
- Motion events under 1.2 s excluded from classification (pre-registered threshold)

Results

Of 1,148 logged events, 41% matched a rostered duty window across the season, with wind, wildlife, and unresolved causes accounting for the rest in a mix that shifted month to month.



Wind-attributed triggers peaked at 39% of May’s logged events, the season’s windiest month on the Bureau’s own gust record.



Total weekly events rose from 34 in week 1 to a season-high 81 in week 11, tracking the onset of the autumn wind season rather than any change to the duty roster.

Discussion & conclusions

The roster accounts for well under half of what the sensor recorded. Wildlife and weather are not noise around the duty schedule; across some months they outnumber it, and the season’s busiest fortnight for the sensor was also its windiest, not its most rostered. Unresolved events held under 10% throughout, suggesting the three-way classification captures most of what the door does. Whether the households read any of this before opening the gate is a separate question the roster alone cannot answer. Next: a possum-specific trigger signature, and a look at whether unrostered households check the coop anyway.

References

1. Meek, P. D., Ballard, G., Fleming, P., Schaefer, M., Williams, W., & Falzon, G. (2014). Camera traps can be heard and seen by animals. *PLoS ONE*, 9(10), e110832. doi:10.1371/journal.pone.0110832
2. Glover-Kapfer, P., Soto-Navarro, C. A., & Wearn, O. R. (2019). Camera-trapping version 3.0: current constraints and future priorities for development. *Remote Sensing in Ecology and Conservation*, 5(3), 209–223. doi:10.1002/rse2.106
3. Hobbs, M. T., & Brehme, C. S. (2017). An improved camera trap for amphibians, reptiles, small mammals, and large invertebrates. *PLOS ONE*, 12(10), e0185026. doi:10.1371/journal.pone.0185026
4. Ostrom, E. (2000). Collective action and the evolution of social norms. *Journal of Economic Perspectives*, 14(3), 137–158. doi:10.1257/jep.14.3.137

Opportunistic wildlife identification only; no possums trapped, tagged, or otherwise inconvenienced by the sensor. We thank the households of the instrumented street.